

Shlok Patel

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EDUCATION

Northeastern University

Boston, MA

Bachelor's Degree in Computer Science and Business Administration (Financial Technology) | GPA: 3.7/4.0 May 2029

Coursework: Software Engineering, Object-Oriented Design, Algorithms & Data Structures, Databases, Discrete Math

Societies: Honors College | Disrupt: Northeastern Fintech Club | Student Government Association | NU ACES | NU Oasis

TECHNICAL SKILLS

Languages: Python, Java, SQL, C++, JavaScript, TypeScript, HTML/CSS

Frameworks & Libraries: React, Next.js, Node.js, Flask, FastAPI, PyTorch, Pandas, NumPy, SciPy, scikit-learn

Developer Tools & Databases: MongoDB, PostgreSQL, MySQL, Supabase, Prisma, Docker, Git, CI/CD, REST APIs

Concepts: Machine Learning, Full Stack, Backend, System Design, Microservices, Database Design, Data Science

EXPERIENCE

Student Government Association

Jan 2026 – Present

Software Engineer

Boston, MA

- Developed a production CMS letting non-technical editors publish to **5,000+** students, cutting publish time by **10%**.
- Architected version history, rollback, role-based access, and soft-delete archiving on a **Next.js/Prisma** stack.
- Engineered **Prisma**-validated API endpoints enforcing schema integrity on every mutation over **PostgreSQL**.
- Shipped features in **Agile/Scrum** sprints, cutting scope ambiguity through structured ticket handoffs with team leads.

Disrupt: The FinTech Initiative

Jan 2026 – Present

Quantitative Analyst

Boston, MA

- Built scalable **Python** back-testing pipelines evaluating trading signals across **10M+** historical data points.
- Developed pairs-trading, mean-reversion, and momentum strategies deployed on **\$50K** of simulated capital.
- Translated equity market research on statistical arbitrage, factor investing, and microstructure into testable signals.

Khoury College of Computer Sciences

May 2026 – Jun 2026

Teaching Assistant, Program Design and Implementation II

Boston, MA

- Led **30+** weekly debugging sessions in **Java**, coaching students on OOP, data structures, and control flow.
- Taught **100+ students** AI-assisted debugging and prompt engineering with **Claude** and **GitHub Copilot**.
- Graded assignments against detailed rubrics with written feedback on SOLID principles, design, and correctness.

Rainfall Learning

Nov 2025 – Jun 2026

Software Lead

Boston, MA

- Led UI development on a real-time IDE with **15+ engineers**, cutting edit conflicts to **2%** via **Yjs** CRDT sync.
- Built a scalable **React** and **TypeScript** UI layer, improving render performance by **20%** during concurrent editing.
- **Dockerized** build pipelines to reduce environment failures by **15%**, hardening deployments with automated testing.
- Engineered **Vitest** and React Testing Library suites, catching UI regressions and improving frontend reliability.

PROJECTS

Derivative Hedging via Reinforcement Learning | Python, PyTorch, Stable-Baselines3, NumPy

Mar 2026

- Built an end-to-end **RL** hedging system over **1,200+ SPY windows** with Greeks and transaction cost modeling.
- Trained **PPO** (500K) and **SAC** (300K) agents, benchmarked against **Black-Scholes** delta over **4,000 episodes**.
- **Tripled the Sharpe ratio** vs. delta hedging, lifting mean P&L from **-0.162** to **+0.064** in high-cost regimes.

Northeastern Dining | React, JavaScript, Node.js, Supabase, PostgreSQL

Nov 2025

- Led a team of 5+ to ship a full-stack dining app end-to-end, owning scope, design, and deployment.
- Built **REST** endpoints with **JWT** auth on a **Supabase/PostgreSQL** backend with sub-second responses.
- Achieved a **40%+** engagement gain through a feedback-driven UI redesign with real-time menus and filters.